



Machine Learning Integrated Biosensor for Disease Detection in Milk Samples

S25-35

Benji Adjepong, Volkan Akkale, Saket
Pendharkar, Jonathan Ravid, Ryan
Recupero, Mason Young

Virginia Tech



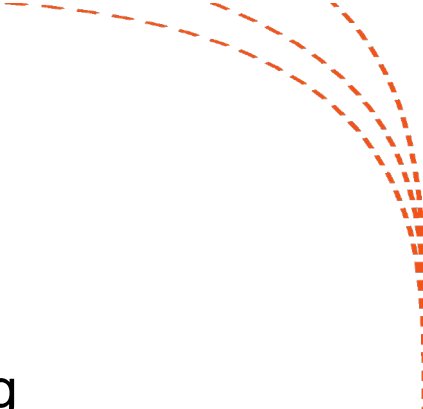
04/23/2025



COLLEGE OF ENGINEERING
BRADLEY DEPARTMENT OF
ELECTRICAL AND COMPUTER ENGINEERING
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Introduction & Objectives



Overview: Our project aims to address the challenges of detecting **Subclinical Mastitis** in dairy cow milk by using biosensing techniques with machine learning.

Develop a dual-modal biosensor system combining **Electrochemical Impedance Spectroscopy** and **Raman Spectroscopy**.

Utilize machine learning algorithms to enhance anomaly detection in milk samples.

Build a software interface and database to support data analysis and remote collaboration.

Design a **portable, scalable, and real-time detection platform** for use in dairy farm environments.

What is Subclinical Mastitis?

SCM is an inflammation in the mammary glands that often does not show any visible signs or symptoms

Often caused by bacterial infection

If left untreated, can develop into clinical mastitis

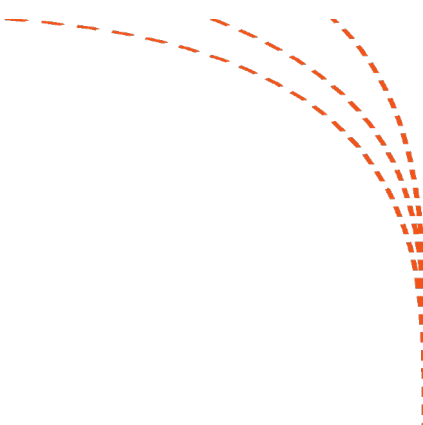
- Abnormal milk or reduction in production
- Fever and in extreme cases: fatality



Mastitic Milk (left) vs
Normal Milk (right) ^[1]



Problem Statement



Subclinical mastitis (SCM) in dairy cows often goes undetected, leading to health issues and financial losses.

Current detection methods are **slow, costly, and impractical** for continuous monitoring.

Our project focuses on the initial design of a system that can utilize **Electrochemical Impedance Spectroscopy (EIS)** and **Raman Spectroscopy** paired with **machine learning** capabilities to detect abnormalities in milk samples.

Proposed Solution

-  Designed and implemented a **machine learning algorithm** to detect anomalies in both optical and electrochemical signals from milk samples.
-  Integrated a **control algorithm** with a **GUI** to enable **autofocus** and **3-axis motorized adjustment**, improving system precision and usability.
-  Built a centralized **database** to store ML predictions, allowing efficient data management and user collaboration.
-  Created a **custom 3D-printed** fixture to securely position the microfluidic device, ensuring stability during testing.

Technical Requirements & Constraints

Required features

- Raman: single capture, continuous monitoring, z-scale, area focus
- Machine learning: python-based system using both data collection methods
- Database: ensuring add functionality

Constraint benchmarks

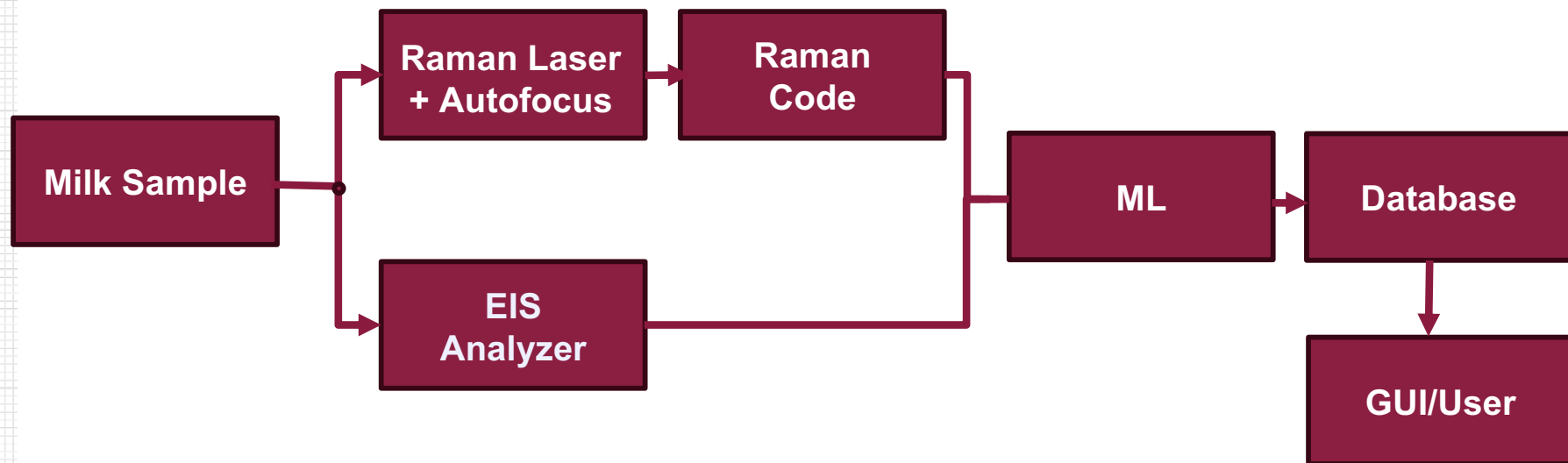
- Raman: Laser must use low operating power, code must be user friendly, rapid focusing
- Machine learning: at least 90% accuracy
- Database: Stay within size guidelines



What makes our product so innovative?

- **ML-Enhanced Detection** allows us to use machine learning algorithms to improve diagnostic precision and reduce false positives.
- **Real-time Results** offer almost instantaneous analysis, giving immediate decision-making
- **Dual-Modal Sensing** lets us combines two detection methods to improve robustness and reliability.

High Level Diagram



How Raman Spectroscopy Works

Monochromatic light from a laser is shined onto the sample.

Most light is scattered elastically (leaves with same energy it enters), but some light is scattered inelastically. This is called Raman scattering.

Spectrometer collects scattered light.

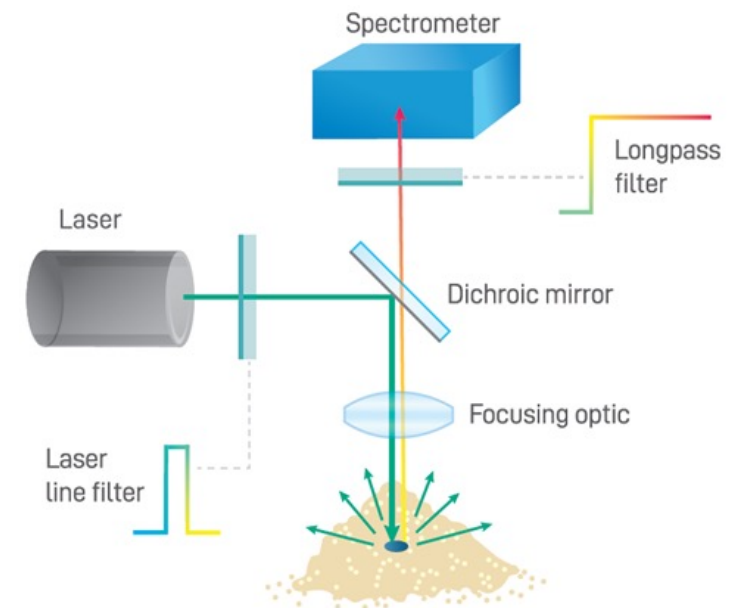
The energy shift of photons is analyzed:

- **Stokes scattering**: photons lose energy
- **Anti-stokes scattering**: photons gain energy

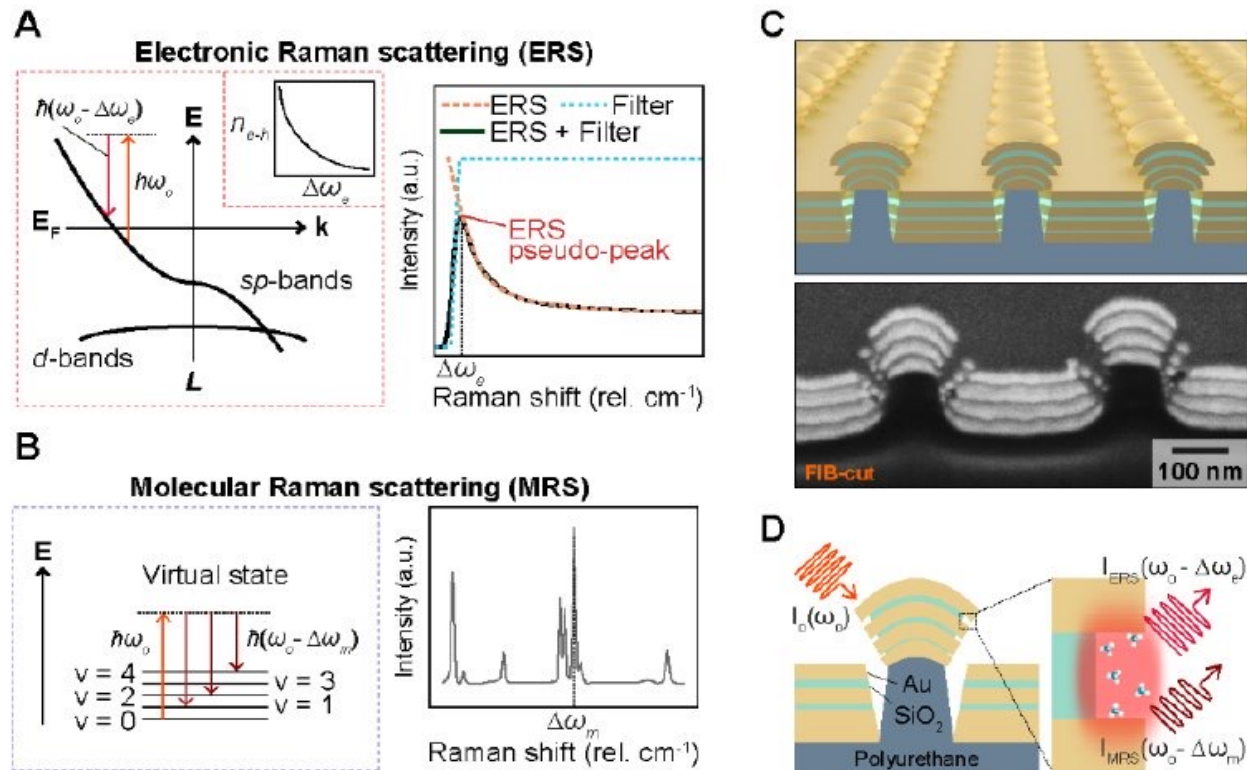
Raman spectrum created based on intensity vs energy shift.

Peaks correspond to specific molecular vibrations.

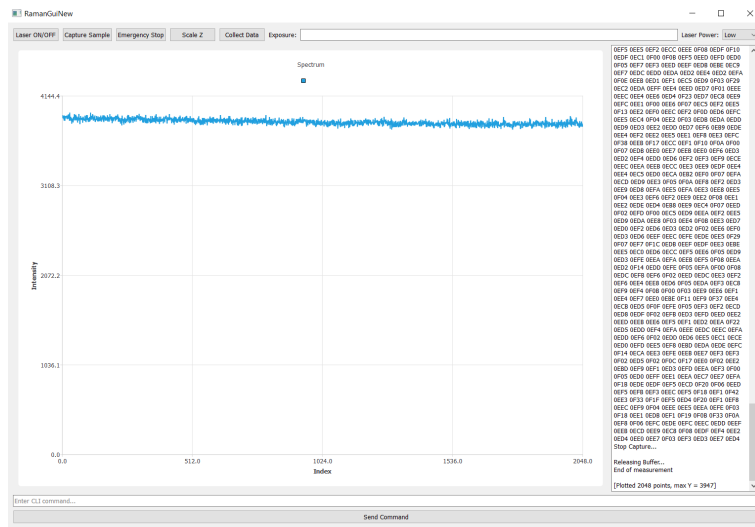
Spectrum is matched to known Raman spectra of molecules.



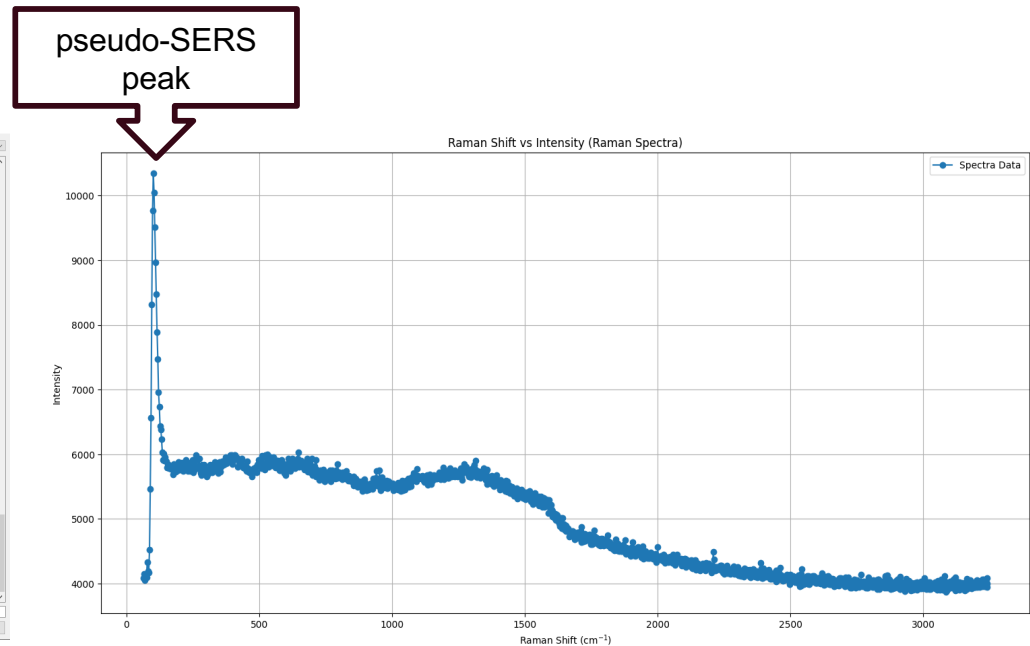
Surface Enhanced Raman Spectroscopy (SERS)



Sample Raman Output



Raman capture on GUI



Capture of SERS nanopillar array device

Autofocus

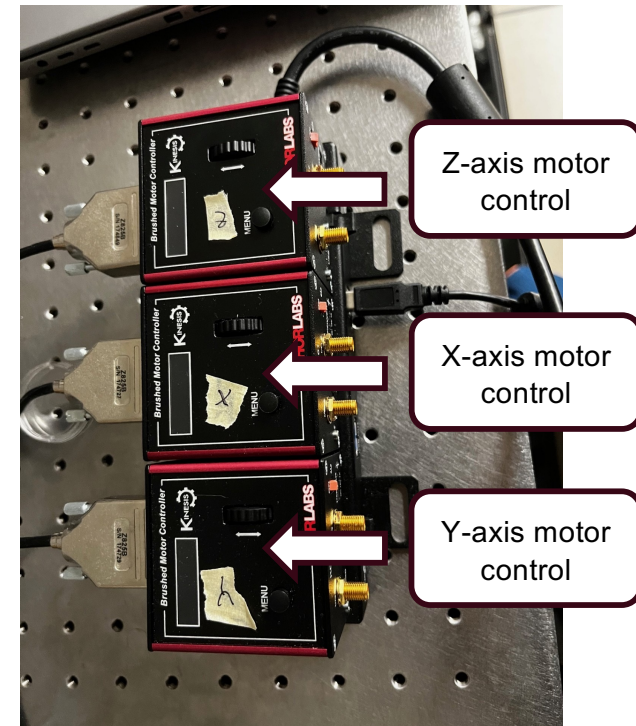
Autofocus simulation

Thorlabs KCube DC Servos

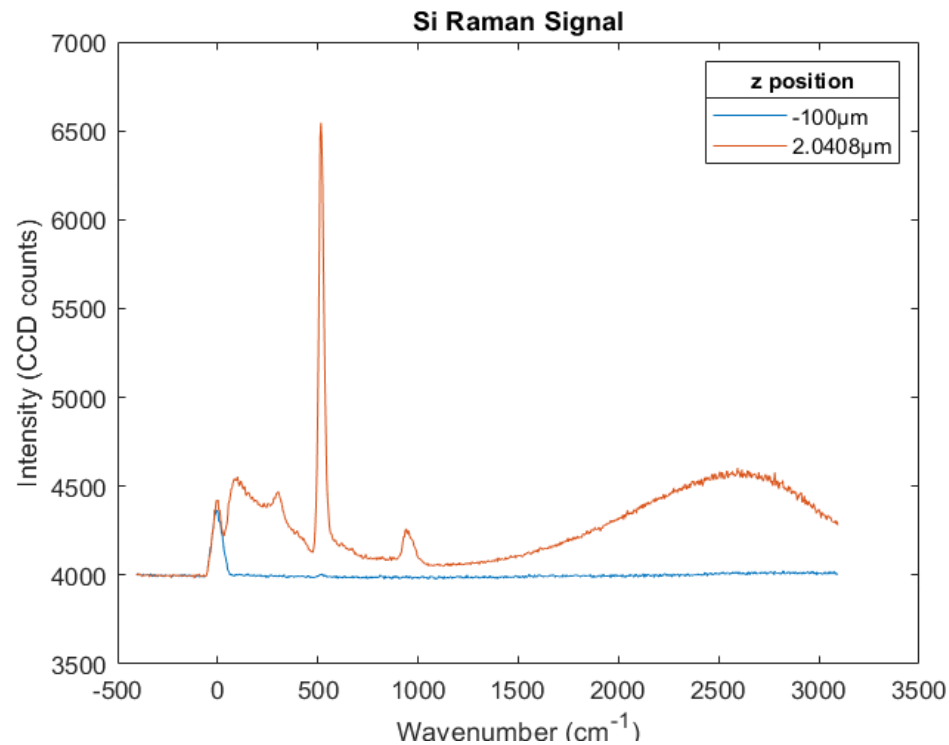
Motors sweep across the z-axis

Location of highest peak in the proper wavelength range is recorded

Motors move the laser to location with highest peak

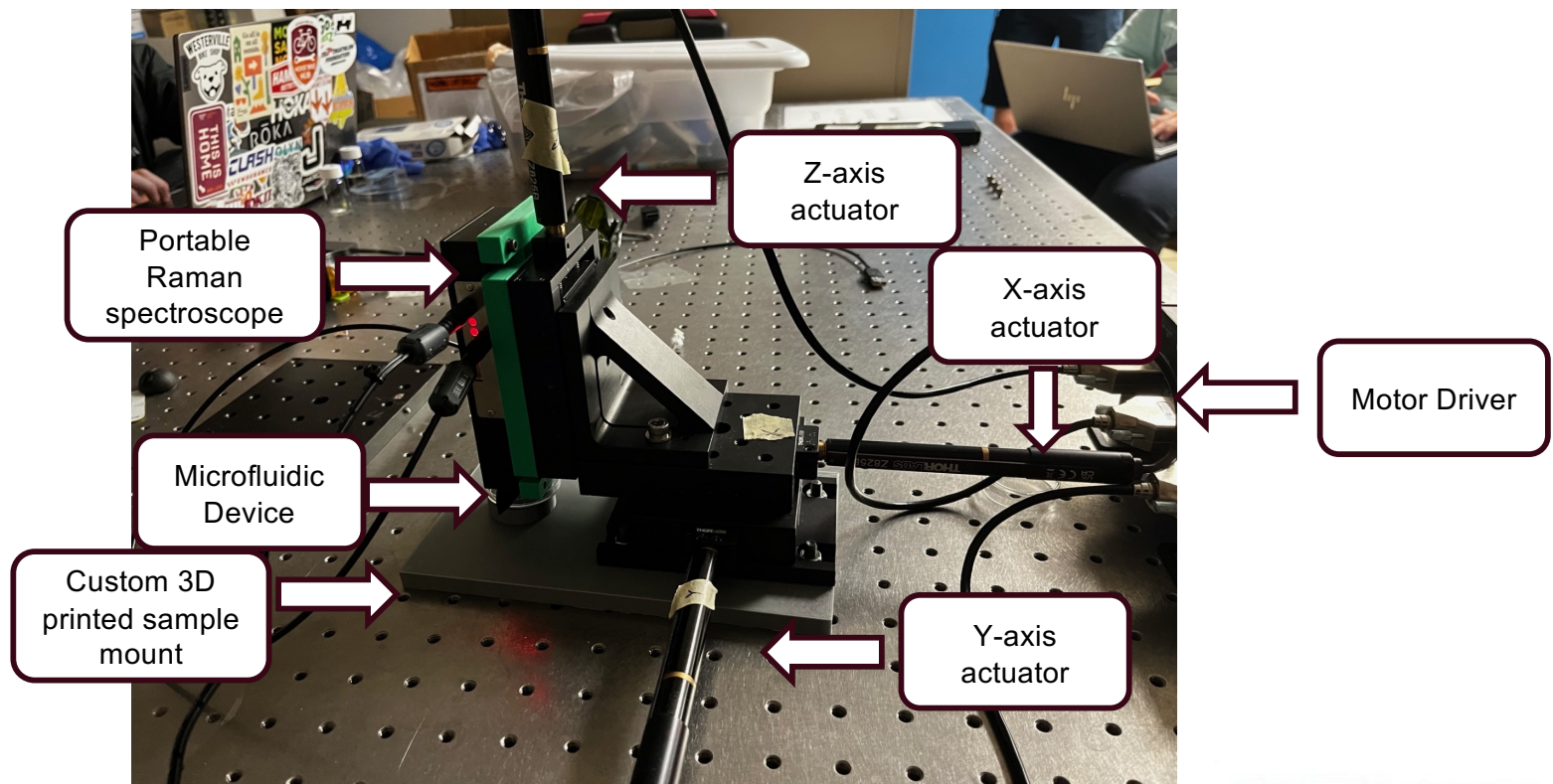


Autofocus Example

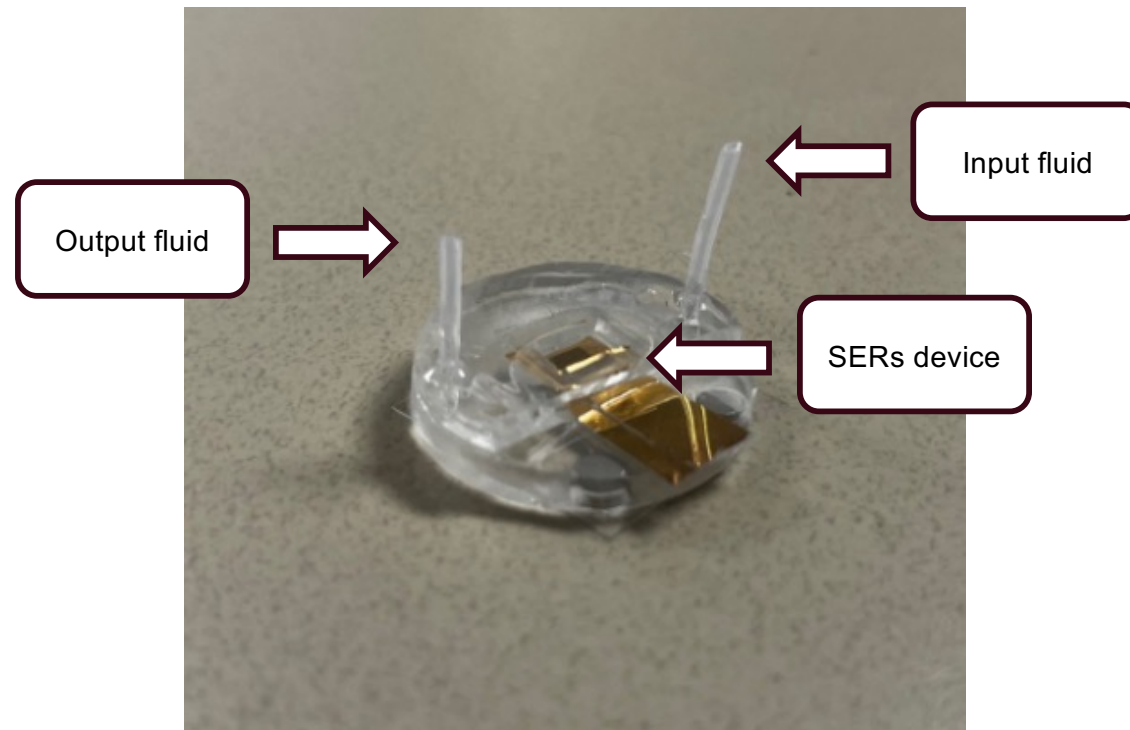


Comparison of Raman spectrum reading of
Silicon **in focus** and **out of focus**

Raman Laser & Motor System



Microfluidic Device

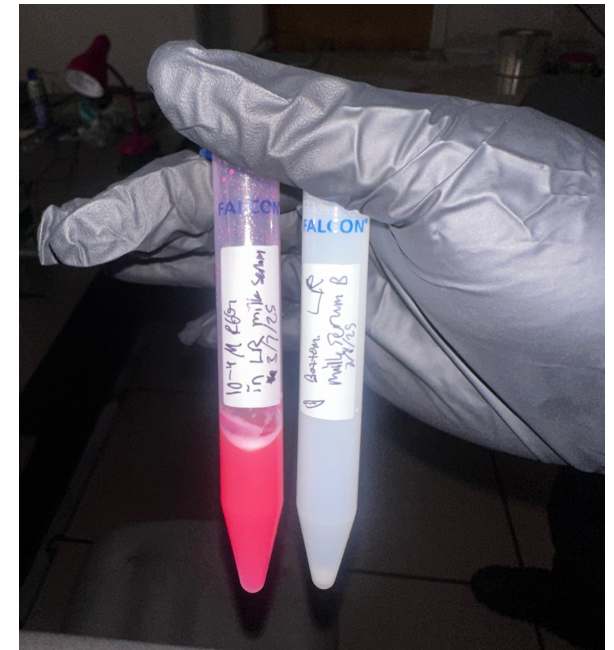


Milk Classification - Milk Samples

We mimicked mastitic milk conditions by treating milk samples to make them distinguishable under Impedance and Raman Spectroscopy

Milk samples were treated with different concentrations of NaCl and EDTA to alter the impedance for classification under spectroscopy

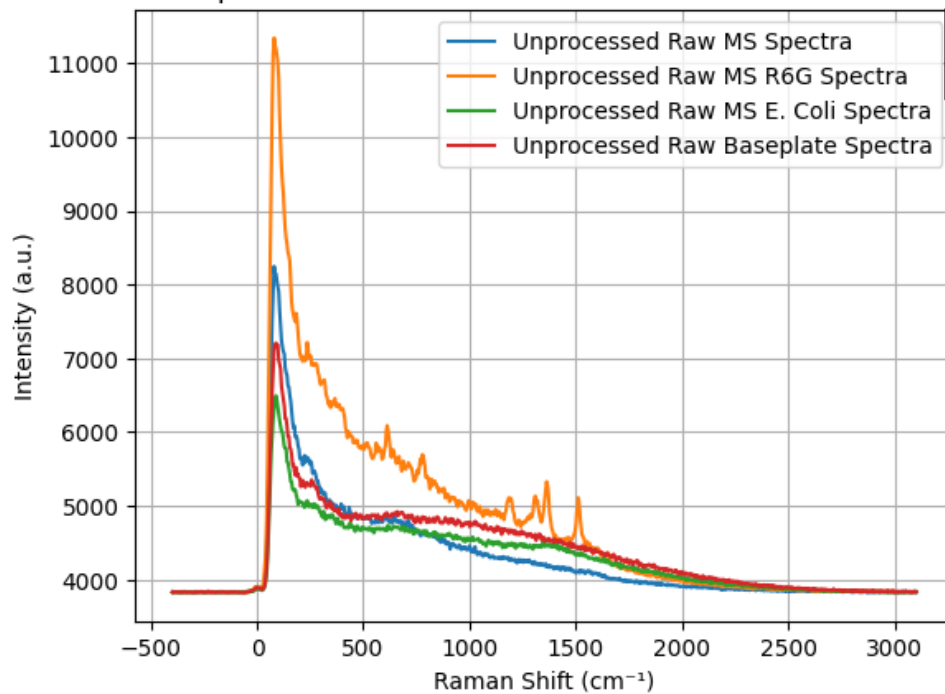
Milk serum samples were treated with 10^{-4} M Rhodamine 6G and lysed E. Coli bacteria to alter Raman signature for multiclass classification



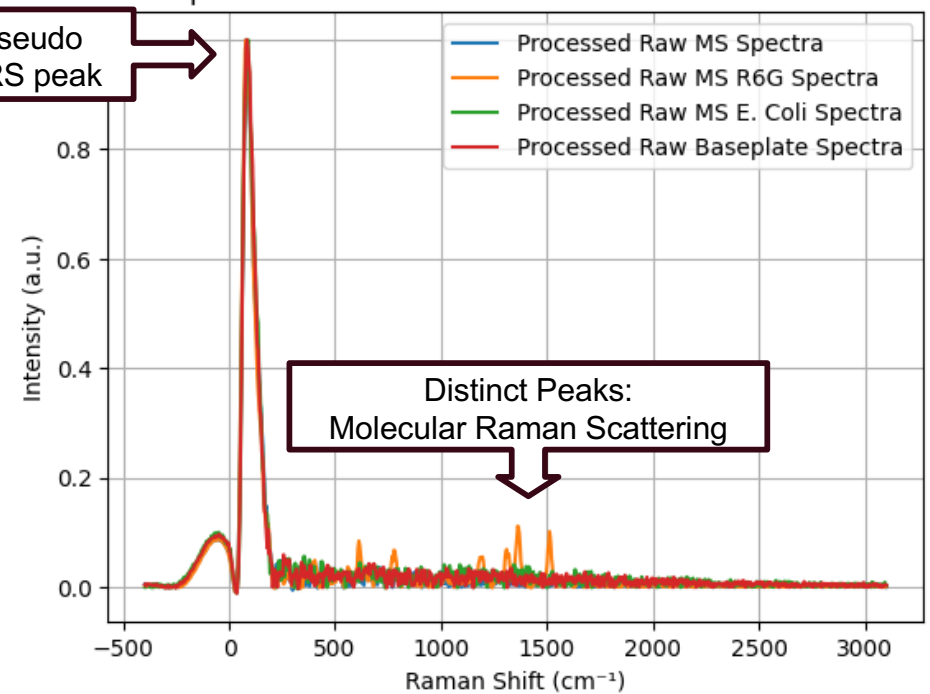
“Rhodamine 6G Treated” vs “Milk Serum”

Milk Classification - Spectra Preprocessing

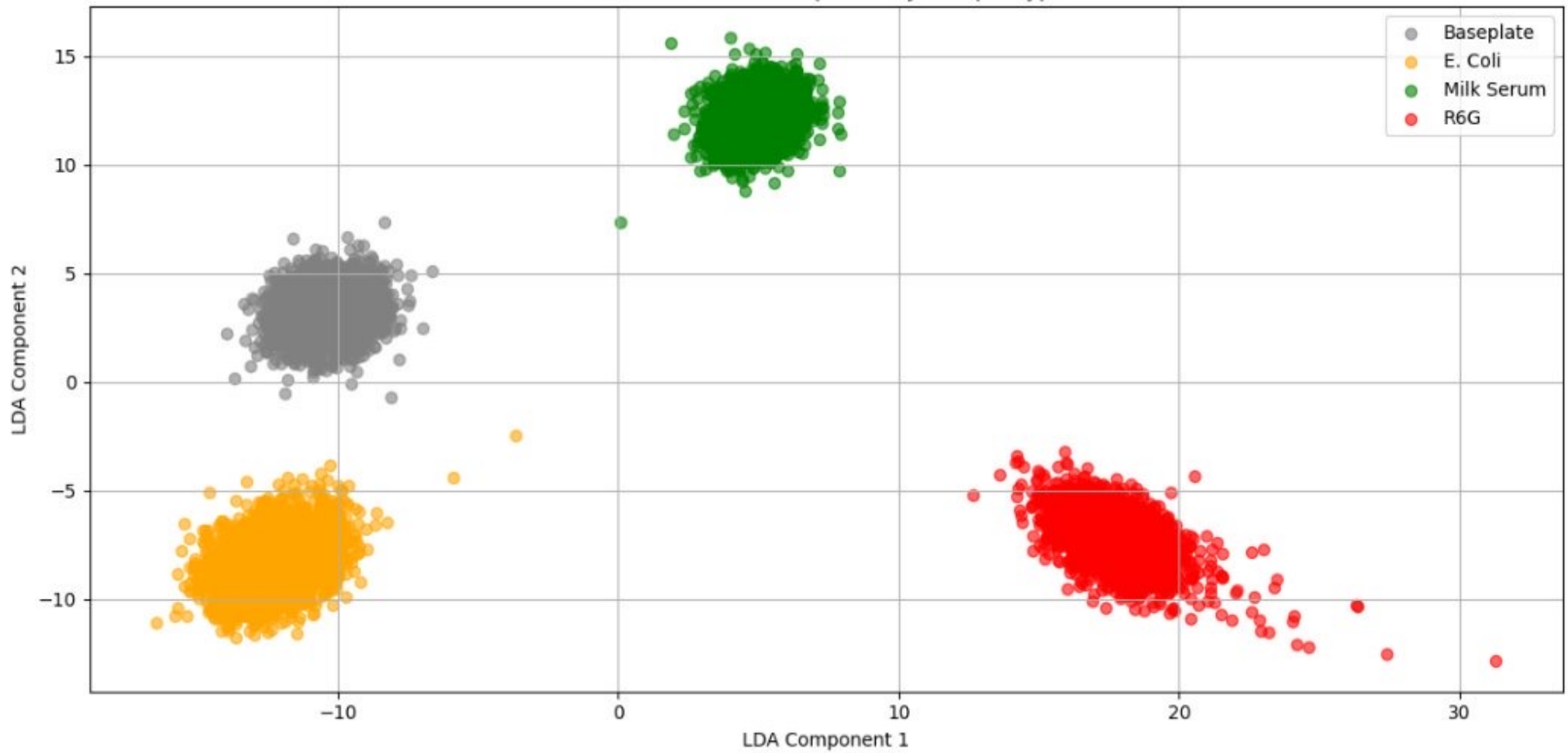
Raman Spectrum Processed: Baseline Correction & Normalization



Raman Spectrum Processed: Baseline Correction & Normalization

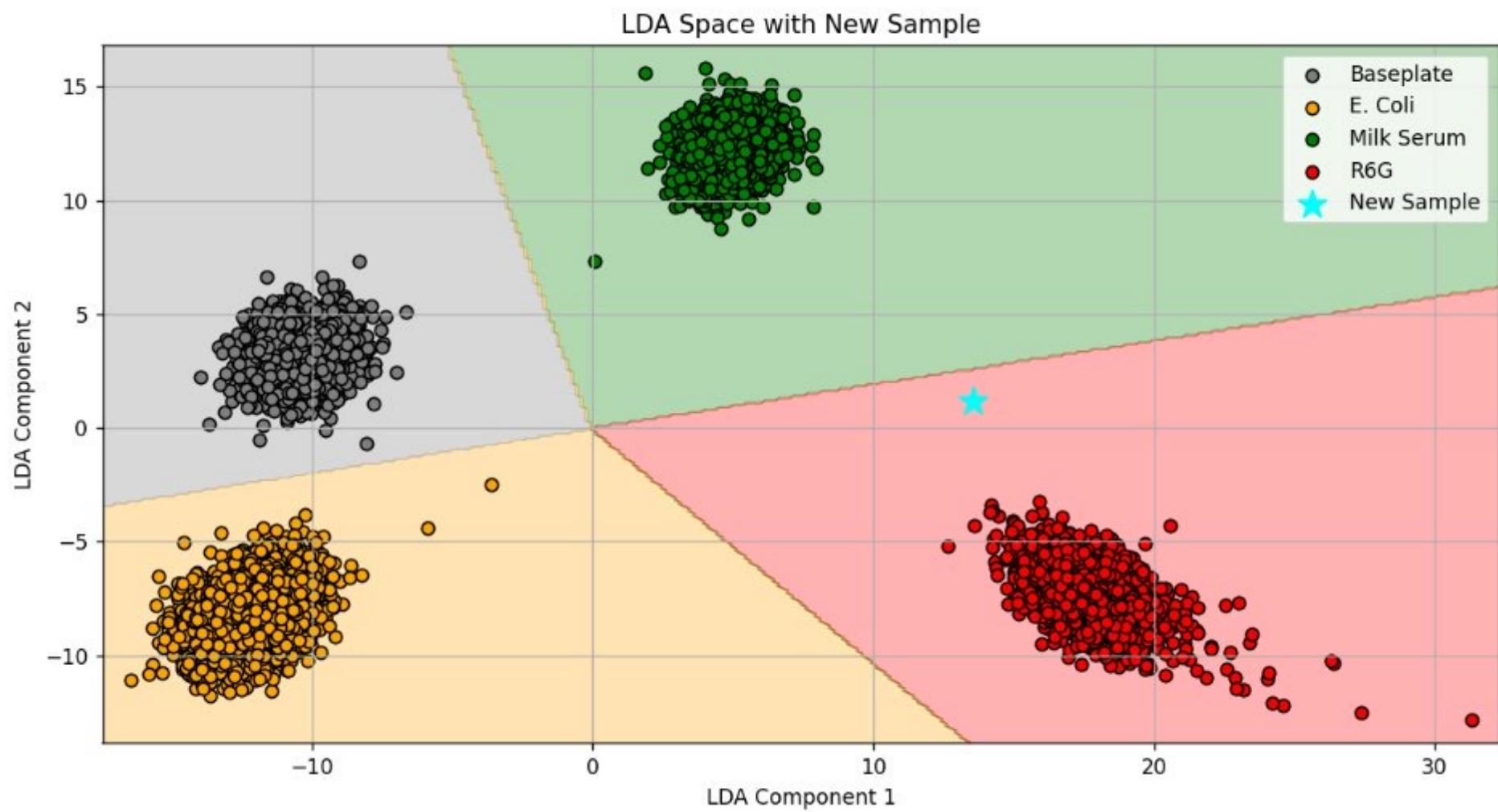


LDA of Milk Raman Spectra by Sample Type



Database Schema - SQLite

results	
id 🔑	integer
timestamp	CURRENT_TIMESTAMP
mastitisDetected	boolean
detectionConfidence	integer





Future Work



Integrate impedance sensor controls in custom software.

Add motor control functionality directly into existing custom software.

Connect microfluidic channel to milking system in a live barn.

Expand data visualization options.

Integrate cloud database functionality.

Further develop ML classification algorithm with more diverse milk datasets.

Port software to other platforms.



Acknowledgements

We'd like to say thank you to the following:

Customers: Dr. Wei Zhou, Dr. Azahar Ali

SME: Dr. Sook Shin

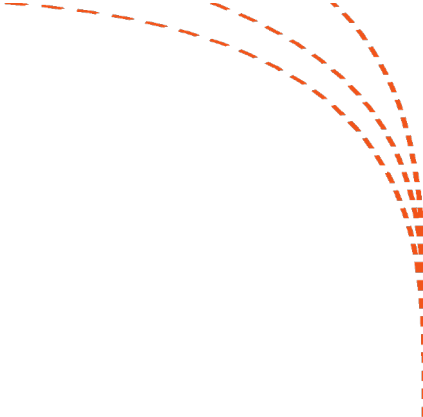
Mentor: Dr. Joe Adams

GRAs: Benjamin Pittelkau, Matin Kachouei



References

[1] ICBF. *Mastitis*[Online]. Available:
<https://www.icbf.com/mastitis/>



Questions?